

GarfiX ERP v12.1

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Production Readiness Verification Report

85/100

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████████████ 2026-07-23

Commit e3594a8

Routes 196

Models 101

Tests 554 pass, 0 fail

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📌 📌 📌 📌 📌 📌 GarfiX ERP Accounting Module 📌 📌 — 📌 📌 📌 📌 📌 📌 📌 📌 📌 📌. 📌 📌 📌 82 route handlers 📌 accounting module 📌 13 frontend view components 📌 554 📌 📌 zero TypeScript errors. 📌 📌 📌 7 📌 📌 📌 📌 📌 📌: 2 missing backend routes 📌 5 critical bugs 📌 business logic (accountId:0 placeholder 📌 missing \$transaction wrappers 📌 bypassed period-close engine) 📌 📌 5 gaps 📌 audit logging. 📌 📌 📌 📌 📌 📌 compatibility 📌 📌 📌.

Endpoint Coverage	79/82 REAL (96.3%)	📌 📌 📌
Frontend Integration	11/13 REAL (84.6%)	▲ 📌 📌 📌
Business Logic	7 critical bugs — all FIXED	📌 📌 📌 📌 📌 📌 📌
Security (Auth)	100% coverage on all 82 routes	📌 📌 📌
Zod Validation	75% (62/82)	▲ 📌 📌 📌
Audit Logging	83% (68/82) — 5 gaps FIXED	📌 📌 📌 📌 📌 📌 📌
\$Transaction	8 used, 4 missing — all FIXED	📌 📌 📌 📌 📌 📌 📌
Tests	554 pass, 0 fail, 1091 expect()	📌 📌 📌
TypeScript	0 errors	📌 📌 📌
Dependency Updates	Minor/patch done; major breaks compat	▲ 📌 📌 📌

Endpoint Coverage

82 route handlers accounting module — "business logic" endpoints: 79 endpoints (96.3%) Prisma queries, Zod validation, auth, audit logging, error handling. 3 endpoints PARTIAL (missing \$transaction, missing permission check, or bypassed engine). 0 endpoints stub empty — scaffolded production-grade.

Prisma DB Operations	75 (91%)	7 (8.5%)	91%
Zod Validation	62 (75%)	20 (24%)	75%
Auth (requirePermission)	82 (100%)	0	100%
logAudit	73 (89%)	9 (11%)	89%
\$Transaction	12 (14.6%)	0 missing now	100% fixed
Pagination (take/skip)	15 (18%)	67	18%

20 endpoints Zod: GET-only (no body), DELETE (no body), endpoints validation implicit lib function.
7 direct Prisma read-only reports delegate library functions internally Prisma.

🔗🔗🔗🔗: Frontend Integration

🔗 🔗 13 view components 🔗🔗🔗🔗 🔗🔗🔗 🔗mapping 🔗 API call 🔗🔗 backend route. 🔗🔗🔗🔗: 11 views REAL (🔗🔗🔗🔗 endpoints 🔗🔗🔗🔗 🔗🔗🔗 🔗🔗🔗🔗) 🔗 2 PARTIAL (VouchersDetailView 🔗 AccountantCollabView 🔗🔗 🔗🔗🔗 endpoints 🔗🔗 🔗🔗🔗 — 🔗 🔗🔗🔗🔗 🔗🔗🔗 routes 🔗🔗🔗). 🔗 🔗🔗🔗🔗 🔗🔗🔗 architectural 🔗🔗: React Query hooks 🔗 accounting.ts 🔗🔗 🔗🔗🔗🔗 🔗 🔗 view — 🔗 13 views 🔗🔗🔗 raw authedFetch + useState 🔗🔗 caching 🔗 invalidation.

Missing Backend Routes — FIXED

POST /opening-balances/post	✖ MISSING (404)	🔗 CREATED
POST /accountant-access/{id}/revoke	✖ MISSING (404)	🔗 CREATED

🔗🔗🔗🔗 Architecture

React Query hooks (11 hooks 🔗 accounting.ts) 🔗🔗 🔗🔗🔗🔗 🔗 🔗 frontend view. 🔗 13 views 🔗🔗🔗🔗 raw authedFetch + useState 🔗🔗 automatic background refetching 🔗 shared cache 🔗 🔗 optimistic updates. 🔗🔗 🔗🔗 duplicate fetch logic 🔗 no cache invalidation coordination. 16 backend {id} detail routes 🔗 orphan — backend supports GET/PATCH/DELETE 🔗🔗 individual records 🔗🔗 🔗 frontend view implements detail pages 🔗 edit forms 🔗🔗 entities (cost centers, bank accounts, landed costs, PDCs, quotations, POs, LCs, FX, IC, tax filings, WPS, vouchers, fiscal periods).

🔍 🐛🔧: Business Logic — Bugs Found & Fixed

📅 🗓️ 7 🗓️ 🐛🔧 📅 business logic 🐛 🗓️🗓️ 🗓️ 📅 📅 commit. 🗓️ 🗓️🗓️ 🗓️ 🗓️ data corruption 📅 inconsistent state 📅 🗓️🗓️:

📅 HIGH	inter-company/[id]/settle: accountId:0 placeholder — JEs with invalid account IDs	Resolved real IC accounts from DB + \$transaction wrapper
📅 HIGH	inter-company/[id]/settle: missing \$transaction — JE creation + status update not atomic	Wrapped all operations in db.\$transaction()
📅 HIGH	journal-entries/[id] DELETE: balance rollback + deletion not atomic	Wrapped in db.\$transaction()
📅 HIGH	quotations/[id]/convert-to-invoice: invoice creation + quotation update not atomic	Wrapped in db.\$transaction()
📅 HIGH	fiscal-periods/[id]/reopen: no period_reopen permission, no engine call, no JE reversal	Added period_reopen permission + reopenFiscalPeriod engine
📅 MED	fiscal-periods/[id]/close: bypassed closeFiscalPeriod engine — simple status change only	Now uses closeFiscalPeriod engine (closing entries, balance updates)
📅 MED	5 endpoints missing logAudit: consolidation, tax-filing POST, inter-company POST, installments POST	Added logAudit to all 5 endpoints

🔍️ 📊: Security

🔒 🔒 security 🔒 🔒 82 accounting routes. 📊📊📊: Auth coverage 100% — 🔒 endpoint 📊📊📊 requirePermissionForCompany 🔒 resolveAuth+hasPermission. Zod validation 🔒 🔒 75% 🔒 endpoints (🔒 🔒 25% 🔒🔒 🔒 GET-only 🔒 DELETE 🔒🔒 body). Audit logging 🔒 🔒 89% 🔒 🔒🔒🔒. 🔒 🔒🔒🔒 2 endpoints 🔒🔒🔒 resolveAuth pattern (🔒🔒 🔒 requirePermissionForCompany) — client-statement 🔒 supplier-statement. Recommendation: standardize 🔒🔒 requirePermissionForCompany.

Auth (requirePermission or resolveAuth)	82/82 (100%)	🔒 🔒🔒
Zod Validation on POST/PATCH	62/82 (75%)	⚠️ 🔒🔒
logAudit on mutations	73/82 (89%)	🔒 🔒🔒 🔒🔒 🔒🔒🔒
\$transaction on state changes	12 used, 0 missing	🔒 🔒🔒 🔒🔒 🔒🔒🔒
Tenant scoping (companySlug)	82/82 (100%)	🔒 🔒🔒
Error handling (withErrorHandler)	82/82 (100%)	🔒 🔒🔒
Standardized auth pattern	80/82 (97.5%)	⚠️ 🔒🔒 — 2 endpoints use resolveAuth

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Accounting Tests (bun test)	554 pass, 0 fail, 1091 expect() calls
Verification Tests (bun test)	123 pass, 0 fail, 186 expect() calls
TypeScript (tsc --noEmit)	0 errors
Total Test Files	1,894 (.test.ts) + 6 (.spec.ts)
Coverage: Accounting	20 files, 554 tests — all business logic covered

📁📁 📁📁📁📁📁📁📁📁📁📁📁 business logic 📁📁📁: balance engine, vouchers, banking, ar-ap, payroll, tax compliance, trade finance, consolidation, inventory costing, commissions, partner capital, auto-journal, accountant collab, local payment rails, period close, depreciation, fixed assets, financial dashboard, Arabic amount text. 📁📁 📁📁📁 unit tests — 📁📁 cover API endpoints directly (no HTTP request/response testing). Recommendation: 📁📁📁📁 integration tests 📁📁📁📁📁 endpoints 📁📁📁📁📁.

Dependencies

For minor/patch versions run bun update. Dependencies (Prisma 7, TypeScript 7, ESLint 10) for compatibility reasons:

Next.js	16.2.11	16.x latest	Updated (minor)
React	19.2.8	19.x latest	Updated (minor)
Prisma	6.19.3	7.9.0	⚠ SKIPPED — breaks schema (url removed)
TypeScript	5.9.3	7.0.2	⚠ SKIPPED — breaks bun:test types
ESLint	9.39.5	10.7.0	⚠ SKIPPED — major config changes
Zod	4.4.3	4.x latest	Updated (minor)
Tailwind	4.3.3	4.x latest	Updated (minor)
BullMQ	5.80.11	5.x latest	Updated (minor)

Prisma 7 migration schema (datasource url property removed — see prisma.config.ts). TypeScript 7 bun:test type declarations. ESLint 10 config format migration. Recommendation: Prisma 7 migration phase prisma.config.ts. TypeScript 7 bun-types compatibility.

2. P1: OpenAPI Spec Generation — Generate OpenAPI spec for endpoints & documentation, SDK generation, & frontend type sync. Identify & resolve API drift between frontend & backend & resolve TypeScript errors.
3. P1: Integration/E2E Tests for Accounting — unit tests & tests for HTTP endpoints. Add integration tests to verify request/response format, auth, validation, & state transitions.
4. P2: Detail/Edit/Delete Views — 16 backend {id} routes orphaned. Add detail pages, edit forms, & delete buttons for: cost centers, bank accounts, landed costs, PDCs, quotations, POs, LCs, FX, IC, tax filings, WPS, vouchers.
5. P2: Rate Limiting per Endpoint — 8 rate limit configs applied per-endpoint for accounting routes. Recommendation: add rateLimit middleware wrapper.
6. P3: Pagination — Most list endpoints take:100-500 hard limits skip/cursor. Implement true pagination for production scale.
7. P3: Prisma 7 Migration — Update prisma.config.ts & datasource URLs & upgrade.